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MsgFusion: Medical Semantic Guided Two-Branch Network for Multimodal Brain Image Fusion

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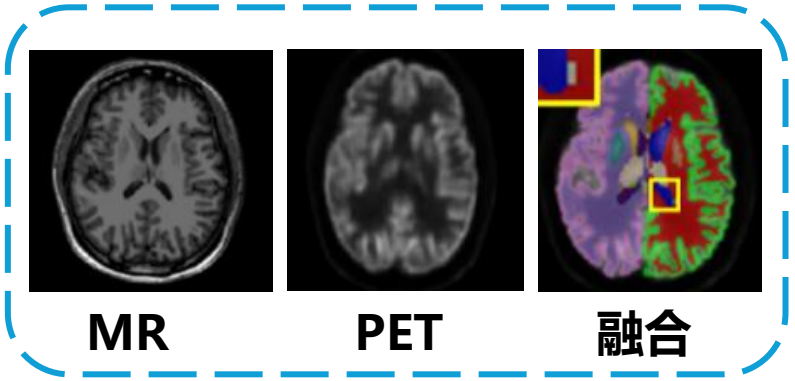
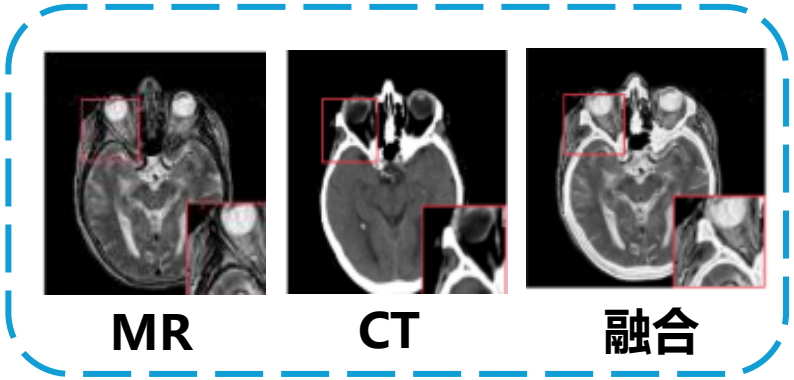
Background

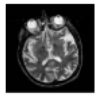
- 医学图像融合在脑疾病诊断中非常重要
- 现有的方法简单套用自然图像的融合策略，忽略不同模态隐藏的独特医学语义信息 (MS-Info)
- 缺乏临床评估，现有的评估指标无法反映融合的结果对医生诊断的实际价值



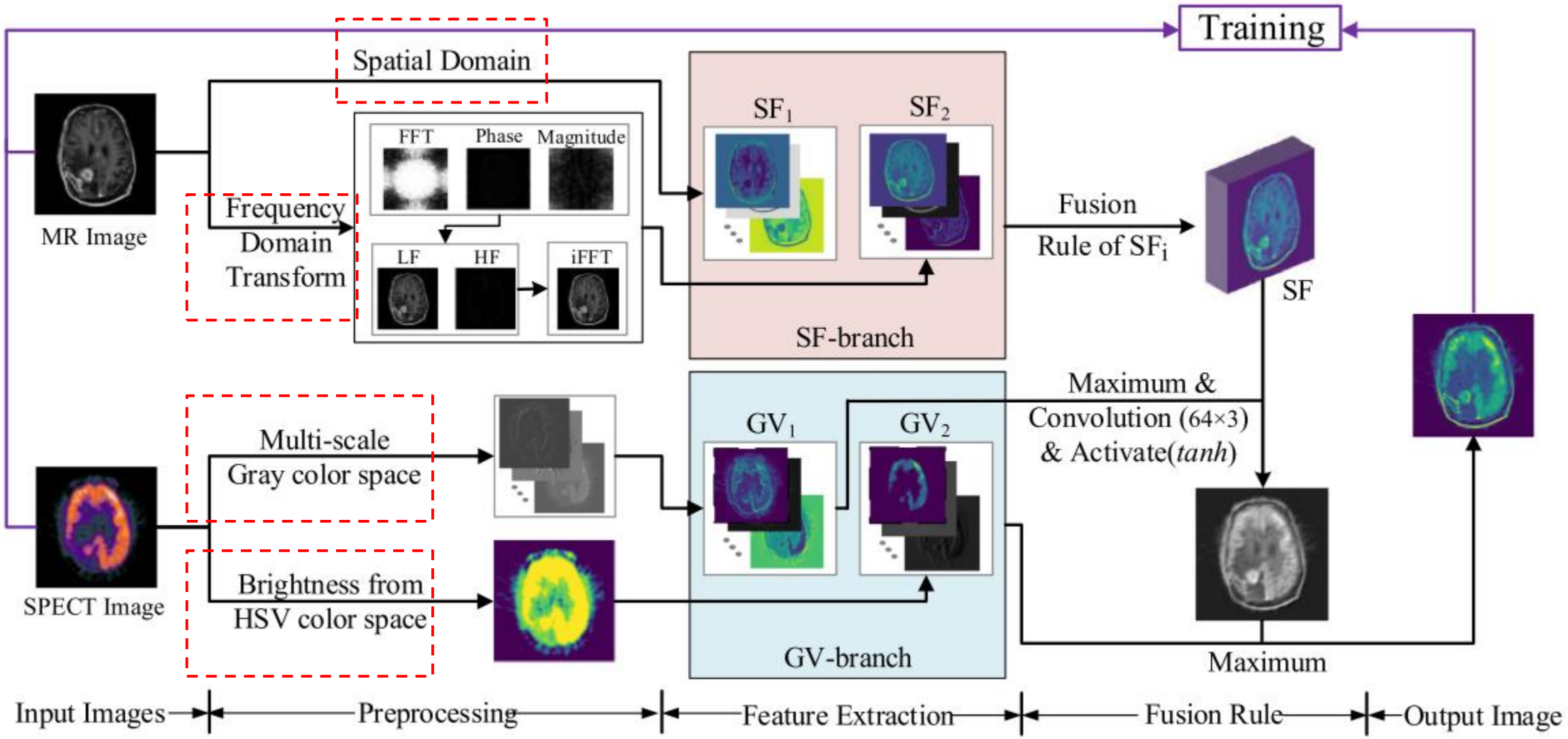


Contribution

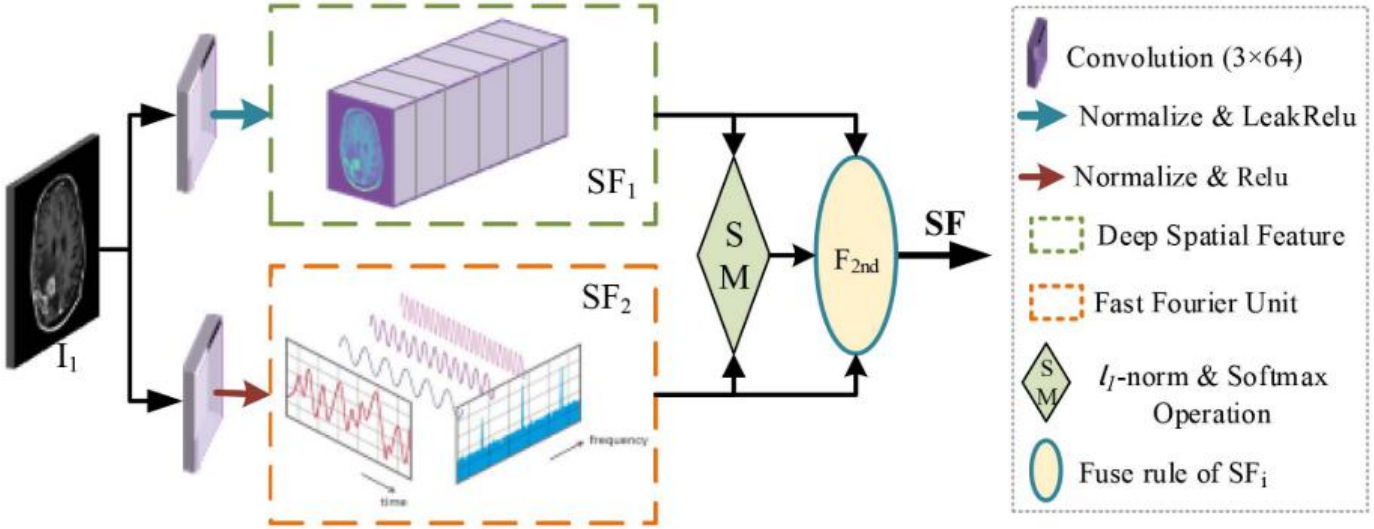
- 本文首次关注多模态医学图像的MS-Info，并将MS-Info映射到对应的图像特征，设计医学语义引导的双分支网络学习不同模态 (MR/CT/PET/SPECT) 的深层特征
- 在SF-branch，首次结合空间域和频率域，用于MR图像的特征提取，既保留了原始图像的特征，又提取MS-Info对应的特征
- 在GV-branch，利用灰度空间特征和HSV颜色空间中的增强亮度 (V') 特征，突出CT图像的特征和PET/SPECT中代谢异常组织的MS-Info
- 最后提出一种基于医生评估的临床评估方法

Modality	Key MS-Info	Image Feature	Extraction Strategy	Branch
MR 	Clear shape of soft tissue	Boundary zone	High frequency band in Frequency Domain	SF-branch
	Clear internal structure of soft tissue	Internal texture detail	Low frequency band in Frequency Domain	
	NIL	More source image information	Spatial Domain	

Overview


Training
Spatial Domain
MR Image
Frequency Domain Transform
FFT
Phase
Magnitude
LF
HF
iFFT
SF1
SF2
SF-branch
Fusion Rule of SFi
SF
Maximum & Convolution (64×3) & Activate(tanh)
Multi-scale Gray color space
Brightness from HSV color space
SPECT Image
GV1
GV2
GV-branch
Maximum
Input Images
Preprocessing
Feature Extraction
Fusion Rule
Output Image

SF-branch



➢ 空间域特征提取 SF_1

使用7×7卷积核（步长1，填充3）将通道数扩增至64，经归一化和激活函数激活后输出空间域特征

➢ 频率域特征提取 SF_2

$$F(u,v)=\frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}f(x,y)e^{-j2\pi(ux/M+vy/N)},$$

二维离散傅里叶变换　空间域 ➡ 频率域

图像大小: $M\times N$　　$f(x,y)$: (x,y)的灰度值　　u,v 是频率变量

$$\sum_{x=0}^{N-1}\sum_{x=0}^{N-1}f(x,y)\cos\left(2\pi\left(\frac{wx}{N}+\frac{vy}{N}\right)\right),$$

实部 Re

$$\sum_{x=0}^{N-1}\sum_{y=0}^{N-1}f(x,y)\sin\left(2\pi\left(\frac{wx}{N}+\frac{vy}{N}\right)\right).$$

虚部 Im

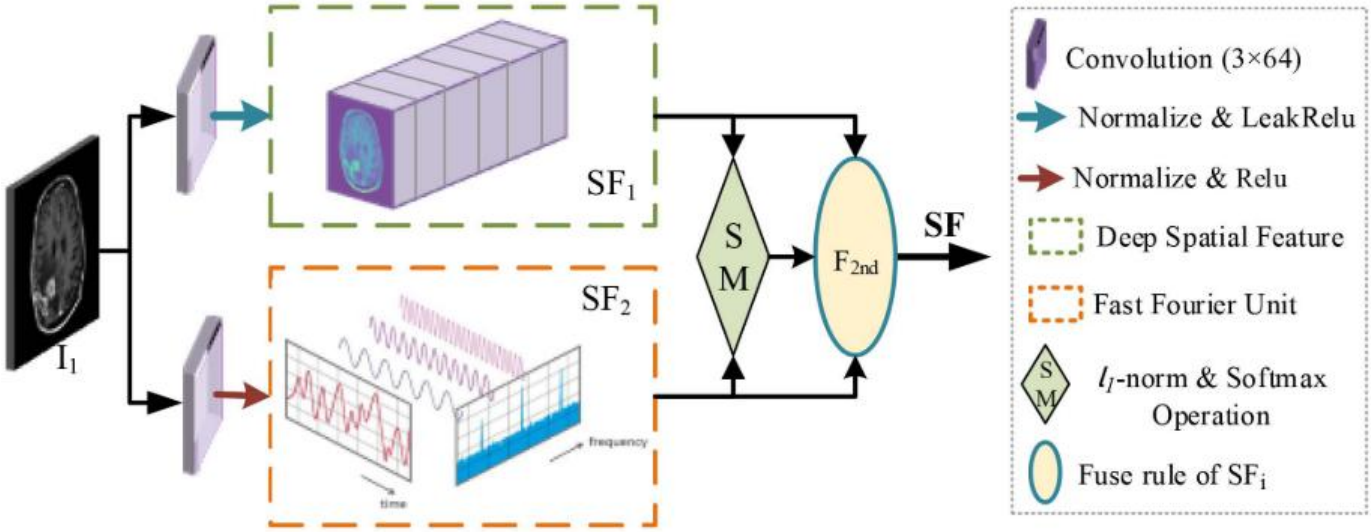
➡ $F(u,v)=Re(u,v)+jIm(u,v)$

SF-branch

$$P(u,v) = |F(u,v)|^2 = Re(u,v)^2 + Im(u,v)^2,$$ 频谱

$$\phi(u,v) = \arctan\left[\frac{Im(u,v)}{Re(u,v)}\right],$$ 相位角

$$|F(u,v)| = \left[Re(u,v)^2 + Im(u,v)^2\right]^{\frac{1}{2}}.$$ 振幅



- 首次使用傅里叶变换进行医学图像融合

图像的振幅包含图像的全局信息，即纹理信息，而相位包含图像的局部信息，即形状

$$\xi_k(x,y) = \frac{\|\psi_k(x,y)\|_1}{\sum_{i=1}^{2}\|\psi_i(x,y)\|_1}, \quad k \in (1,2)$$

$$\varphi = \sum_{i=1}^{2} \xi_{k(x,y)}^{(i)} \times \psi_i.$$

(x,y) : 表示特征图的坐标

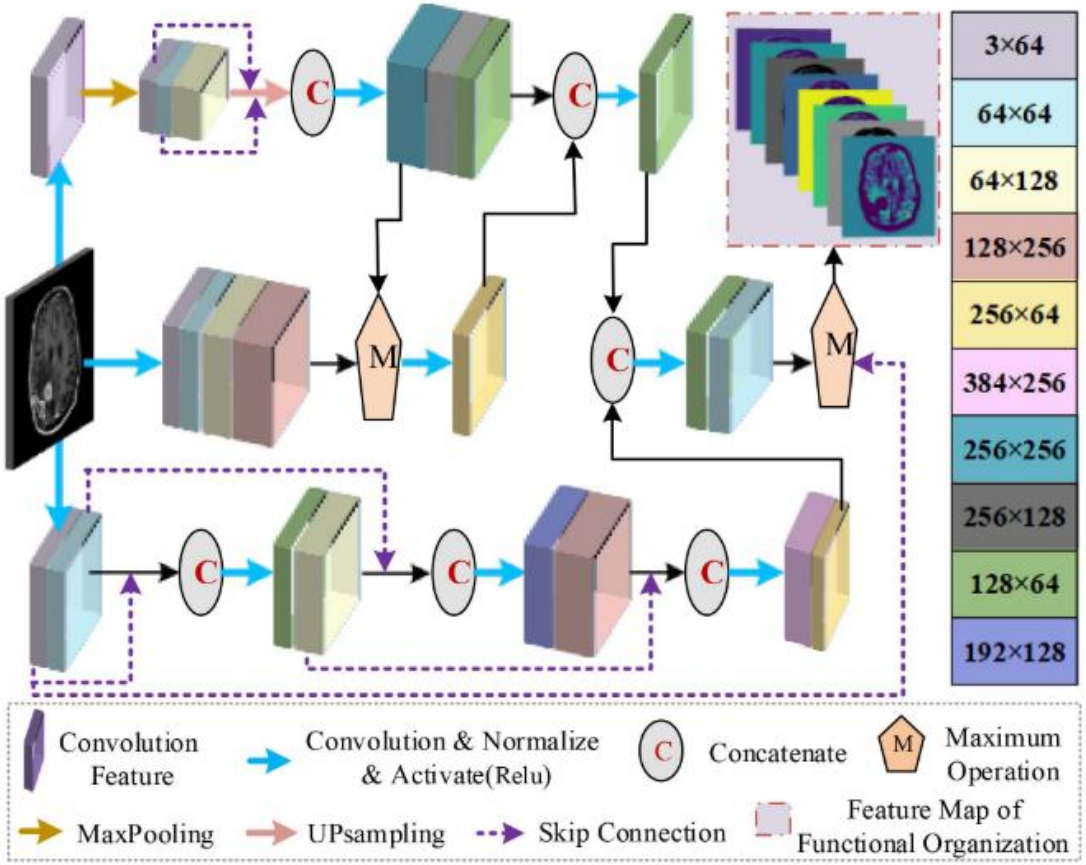
φ 融合特征图：表示两个增强特征图的叠加

GV-branch

➢ 利用多尺度的策略提取深层卷积特征

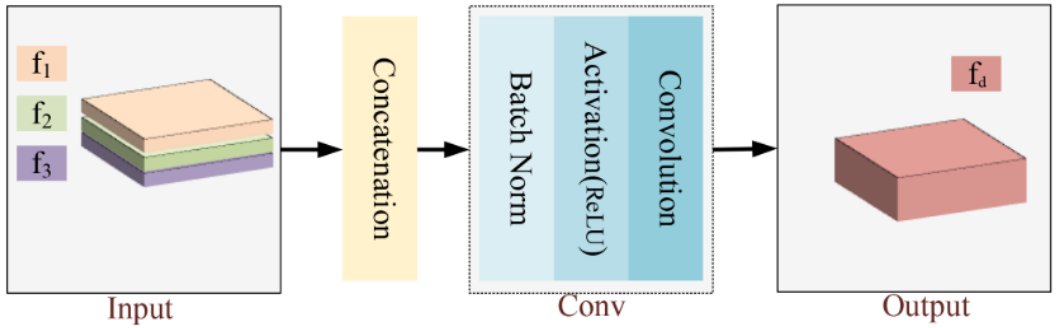
$$f_d = H_d([f_0, f_1, \ldots, f_{d-1}]),$$

Concatenate、BN、Activation和Convolution



➢ 跳跃连接

捕捉跨区域的长期和多层依赖关系，减少特征转移过程中的损失



GV-branch

➢ 通过颜色空间变换获得亮度信息

RGB分别表示红色(R)、绿色(G)和蓝色(B)，是日常生活中接触比较多的颜色空间，比如彩色的图像等

HSV是1978提出的一种颜色空间，由色度(Hue)、饱和度(Saturation)和亮度(Value)组成

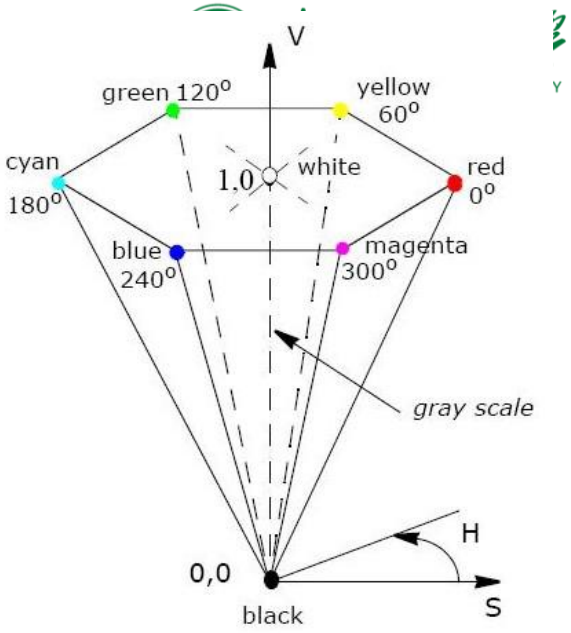
$$H = \begin{cases} 0^\circ & \text{if } M = m \\ 60^\circ \times \frac{G-B}{M-m} + 0^\circ, & \text{if } M = R \text{ and } G \geq B \\ 60^\circ \times \frac{G-B}{M-m} + 360^\circ, & \text{if } M = R \text{ and } G < B \\ 60^\circ \times \frac{B-R}{M-m} + 120^\circ, & \text{if } M = G \\ 60^\circ \times \frac{R-G}{M-m} + 240^\circ, & \text{if } M = B \end{cases}$$

$$S = \begin{cases} 0, & \text{if } M = 0 \\ \frac{M-m}{M} = 1 - \frac{m}{M}, & \text{otherwise} \end{cases}$$

$$V = M.$$

$$M \in Max(R,G,B)$$

$$m \in Min(R,G,B)$$



H $0^\circ \longrightarrow 360^\circ$

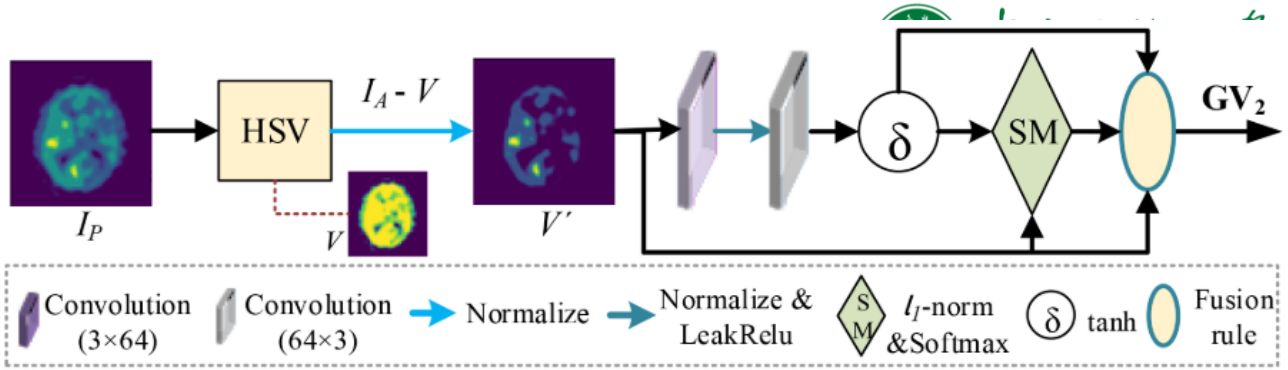
S $0 \longrightarrow 1$

V 0(暗) $\longrightarrow$ 1(明)

本文采用了自定义的V

GV-branch

I_p 原始PET图像　I_A PET图像



Algorithm 1: New Luminance Computation

Input: RGBA image $I_p : (R, G, B, A)$
Output: New luminance value V'

newLum (I_p):
　$I_A \leftarrow$ Compress (I_p);
　$I_{HSV} : (H, S, V) \leftarrow$ Conver (I_p);
　$V' \leftarrow I_A - V$;
　return V';
Compress (I_p):
　$I_A \leftarrow I_p : (R) + I_p : (G) + I_p : (B)$;
　return I_A;
Conver (I_p):
　$I_A \leftarrow$ Compress (I_p);
　$I_{HSV} \xleftarrow{Eq.(1)} I_A$;
　return $I_{HSV} : (H, S, V)$;

原始的V 表示的区域不够清晰，无法准确定位病变

计算新的亮度信息V'

Fusion and Reconstruction

首先对结构特征(SF和GV1)进行融合，然后将融合结果与功能特征(GV2)进行融合，有利于保留和增强待融合医学图像的关键MS-Info。

- Loss Function

$$L = \omega L_S + (1-\omega) L_P.$$

结构相似性　　像素损失

$$L_S = \sum_{i=1}^{2} \alpha_i \left(1 - \frac{\left(2\mu_{I_f}\mu_{I_i} + c_1\right)\left(2\sigma_{I_f}\sigma_{I_i} + c_2\right)}{\left(\mu_{I_f}^2 + \mu_{I_i}^2 + c_1\right)\left(\sigma_{I_f}^2 + \sigma_{I_i}^2 + c_2\right)} \right)$$

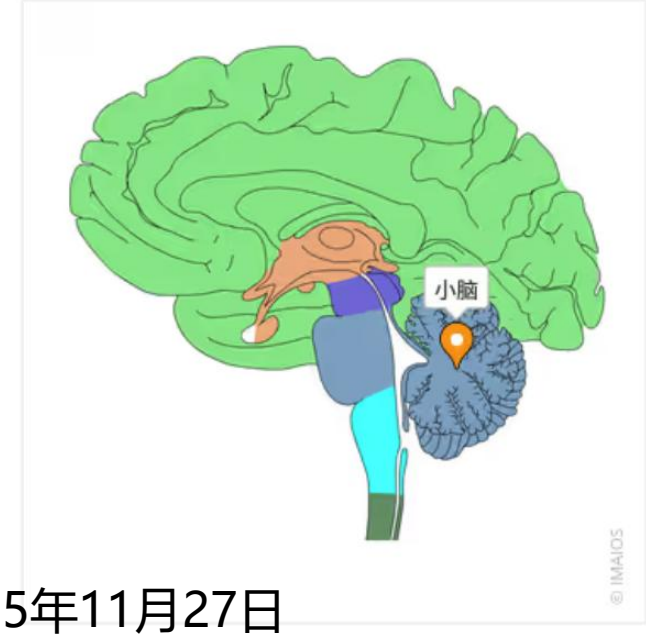
I_f 融合的结果，　$I_i, i = 1,2$ 输入图像和源图像

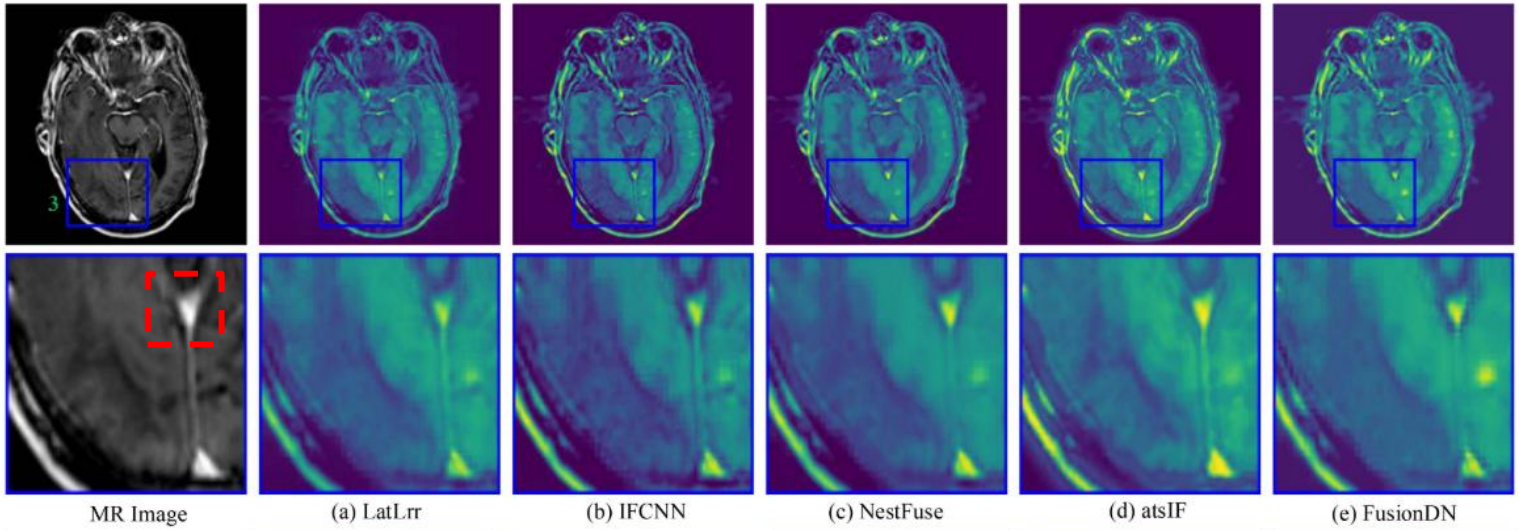
$$L_P = \beta \sum_{j=1}^{n} \left(I_{f_j} - I_{1_j}\right)^2 + (1-\beta) \sum_{j=1}^{n} \left(I_{f_j} - I_{2_j}\right)^2$$

Experiments

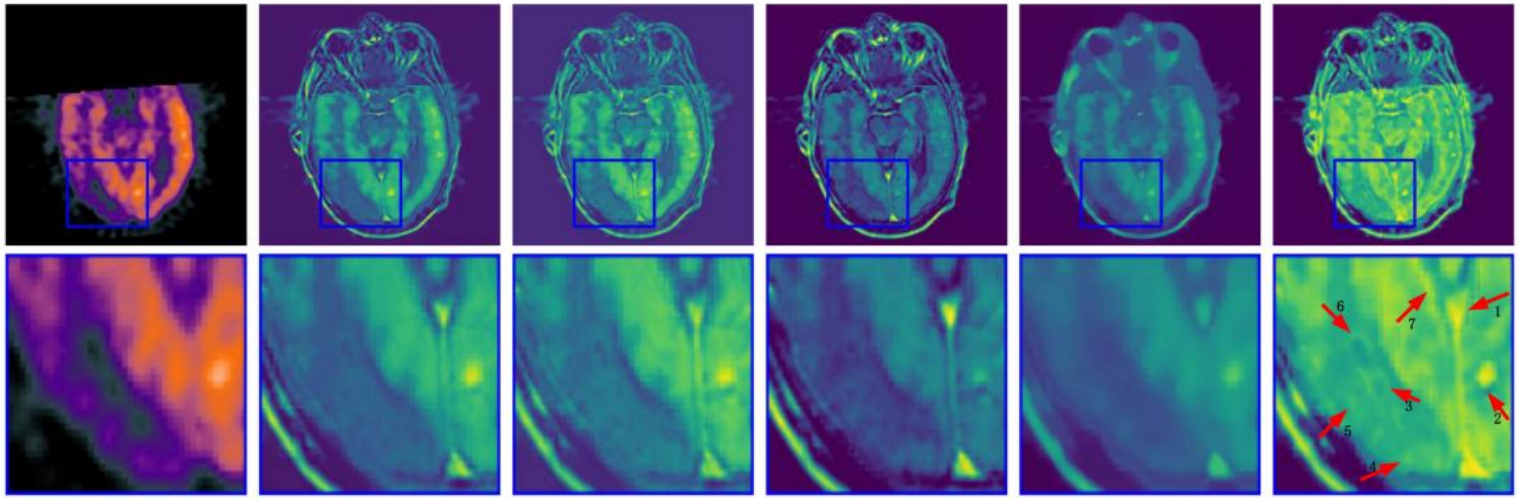
➢ Fusion of MR-SPECT Pairs

区域1是小脑结节。在MR源图像中可以看到轮廓清晰的小白色三角形，右上角略有模糊；在SPECT源图像中，没有明显的特征。其他的融合结果中，图像显得模糊，与周围组织难以区分。





MR Image (a) LatLrr (b) IFCNN (c) NestFuse (d) atsIF (e) FusionDN

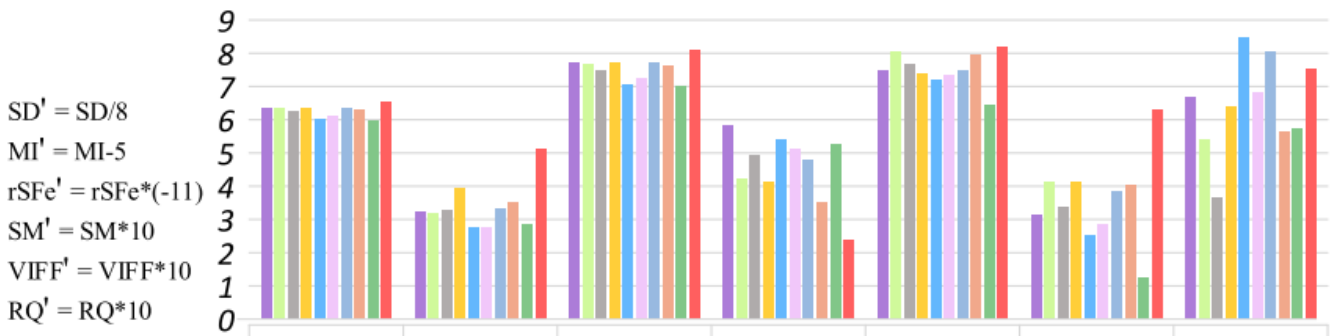


SPECT Image (f) FusionGAN (g) FunFuseAn (h) WPADCPCNN (i) OLTPSpS (j) MsgFusion

Experiments

➢ Fusion of MR-SPECT Pairs

本文的MsgFusion在6个评估指标中处于领先

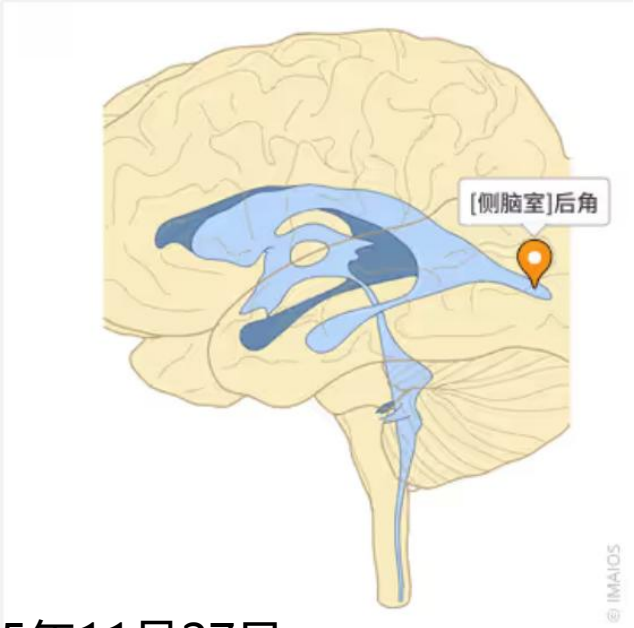


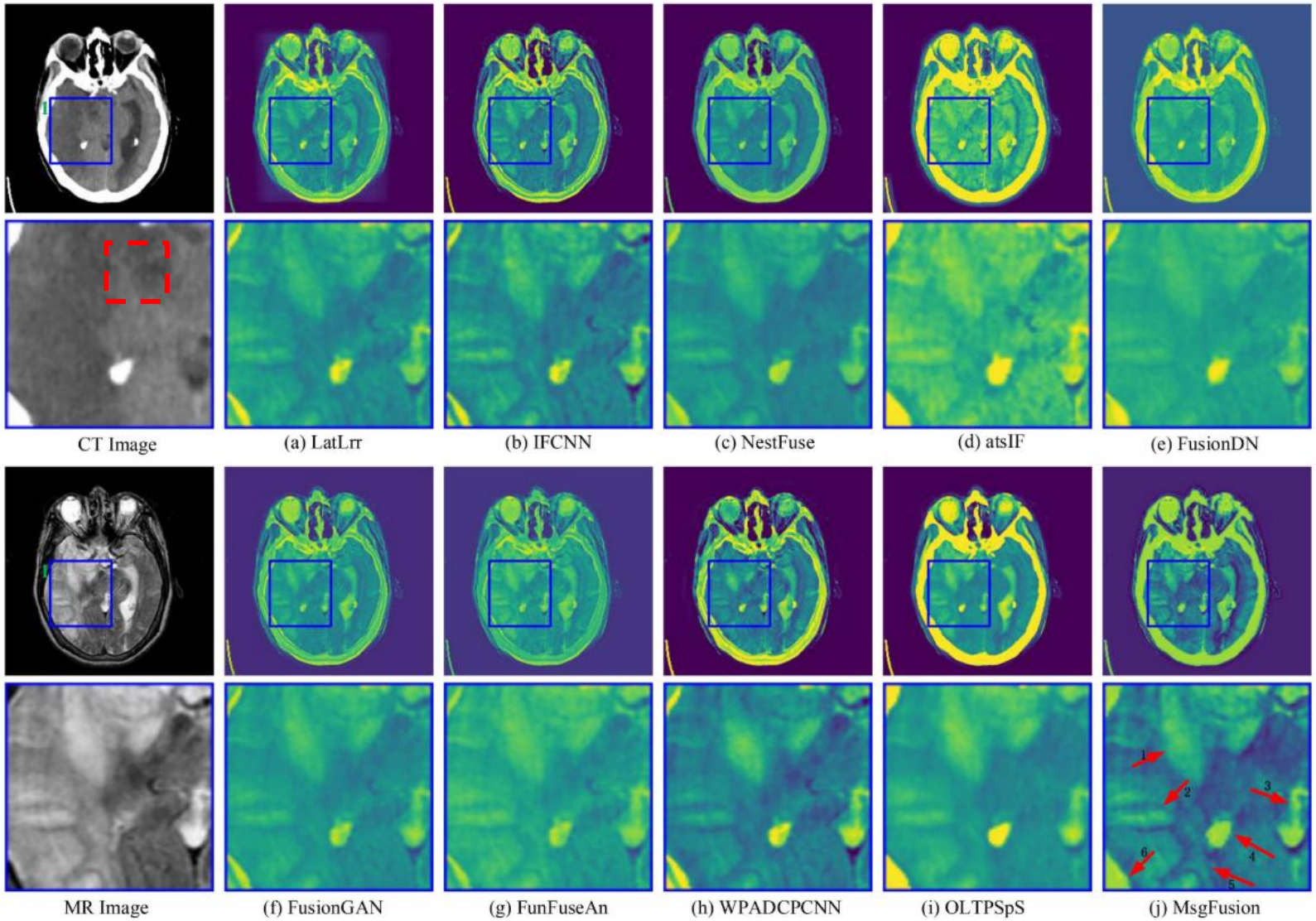
	EN	SD'	MI'	rSFe'	SM'	VIFF'	RQ'
LatLrr	6.371	3.2680	7.7421	5.8223	7.504	3.161	6.697
IFCNN	6.3416	3.2103	7.6833	4.2163	8.06	4.126	5.438
NestFuse	6.2592	3.3050	7.5184	4.9511	7.669	3.413	3.662
atsIF	6.3732	3.9762	7.7464	4.1503	7.403	4.125	6.428
FusionDN	6.0335	2.7899	7.067	5.4043	7.215	2.541	**8.488**
FusionGAN	6.1349	2.7832	7.2698	5.1337	7.335	2.873	6.844
FunFuseAn	6.3731	3.3644	7.7462	4.7839	7.492	3.876	8.073
WPADCPCNN	6.3289	3.5269	7.6578	3.5178	7.956	4.033	5.635
OLTPSpS	6.0088	2.8887	7.0177	5.2987	6.468	1.28	5.550
MsgFusion	**6.5606**	**5.1588**	**8.1212**	**2.3991**	**8.218**	**6.302**	7.528

Experiments

➢ Fusion of MR-CT Pairs

区域4为侧脑室后角，主要体现在CT图像上。在MR图像上不是特别明显，但可以通过与周围组织的仔细区分来发现

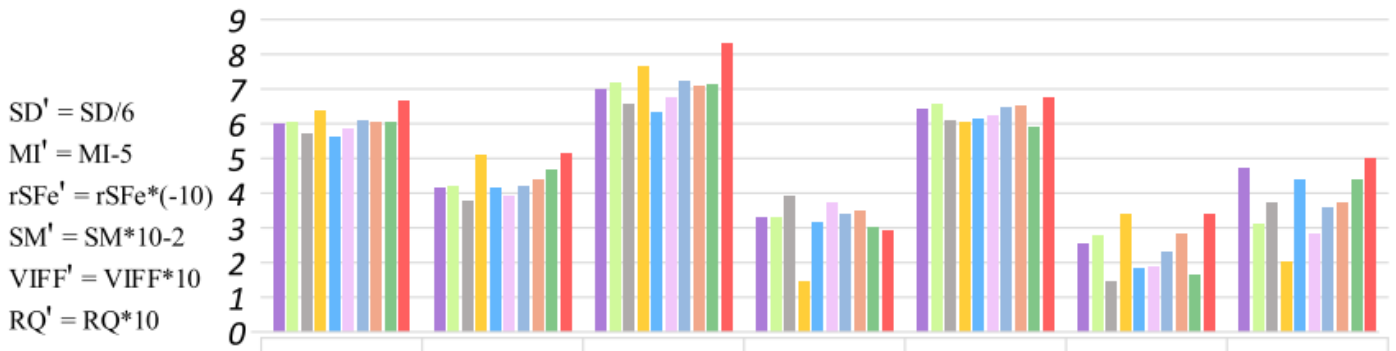




Experiments

➢ Fusion of MR-CT Pairs

本文的MsgFusion在6个评估指标中处于领先



	EN	SD'	MI'	rSFe'	SM'	VIFF'	RQ'
LatLrr	5.9757	4.1380	6.9811	3.323	6.427	2.527	4.707
IFCNN	6.0634	4.1904	7.1725	3.312	6.586	2.769	3.089
NestFuse	5.7155	3.7843	6.5621	3.939	6.099	1.478	3.735
atsIF	6.3552	5.0766	7.6621	**1.471**	6.036	3.373	2.028
FusionDN	5.6086	4.1469	6.3117	3.153	6.132	1.846	4.397
FusionGAN	5.8469	3.9004	6.7672	3.718	6.24	1.865	2.825
FunFuseAn	6.0964	4.1793	7.2422	3.396	6.448	2.316	3.599
WPADCPCNN	6.0492	4.3775	7.0985	3.499	6.506	2.825	3.747
OLTPSpS	6.0665	4.6828	7.1331	3.011	5.888	1.649	4.401
MsgFusion	**6.6525**	**5.1458**	**8.3049**	2.933	**6.756**	**3.415**	**5.009**

Experiments

➢ Ablation Study

Whole image	EN	SD	MI	rSFe	SM	VIFF
w/o FFU	4.5027	56.8141	9.0054	-0.5018	0.2704	0.3219
w/o LightV	3.5768	36.7745	7.1537	-0.2977	0.3523	0.1262
MsgFusion	4.476	67.5634	8.952	-0.4103	0.3205	0.4076
Average ROIs	EN	SD	MI	rSFe	SM	VIFF
w/o FFU	6.9201	52.0514	13.8401	-0.4204	0.5594	0.3227
w/o LightV	5.9916	34.0278	11.9832	-0.3802	0.6118	0.1738
MsgFusion	6.9598	54.9381	13.9195	-0.3467	0.5830	0.3652

MsgFusion在Whole image和Average ROIs中都取得了最好的评估指标

Experiments

➢ Questionnaire Survey

	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9	Q_{10}	Q_{11}	Q_{12}	Q_{13}	Q_{14}	Q_{15}	$\sum$
LatLrr [7]	6	12	9	5	3	3	7	4	8	4	7	4	10	6	10	98
NestFuse [17]	16	10	7	9	8	10	8	6	7	1	6	5	5	6	2	106
atsIF [8]	9	7	7	5	10	2	9	8	12	5	9	15	3	10	4	115
FusionGAN [15]	7	8	7	0	6	4	2	10	4	15	3	2	11	7	2	88
FunFuseAn [14]	0	1	1	6	4	4	5	4	9	7	4	6	6	1	12	70
Our MsgFusion	7	5	12	17	12	21	13	12	4	11	14	10	9	13	11	172

MsgFusion在15个问题中，得到了172个计数选择，远超过次优方法atsIF的115

Experiments

➢ Computational Cost

LatLrr	IFCNN	NestFuse	atsIF	FusionDN
41.2933	0.2241	0.9762	6.7982	0.6873
FusionGAN	FunFuseAn	WPADCPCNN	OLTPSpS	MsgFusion
0.4924	0.3599	39.6705	48.59	2.0048

MsgFusion的平均运行时间是2.0048秒，虽然不是最优的方法，但总体上来说可以接受的

Thank you